

ATF COOLING SYSTEM

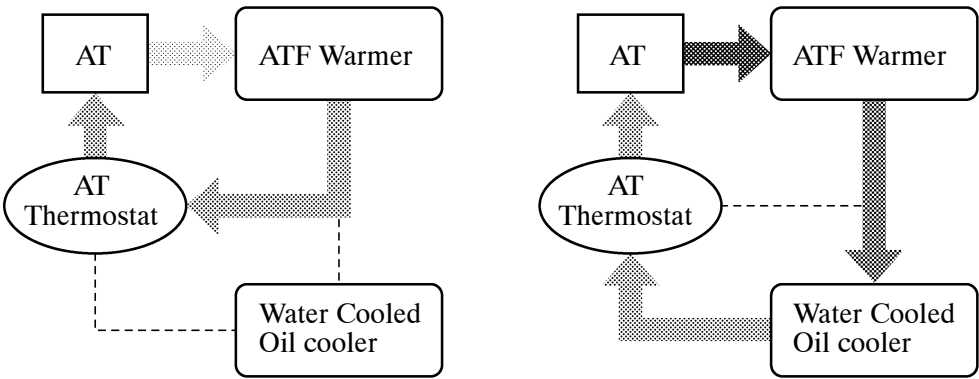
1. General

This automatic transmission uses an ATF warmer that accelerates a rise in the ATF temperature, and an ATF cooler that cools down ATF. The settings of an ATF warmer, water cooled oil cooler, and air cooled vary in accordance with the destinations.

○: Standard —: Not available

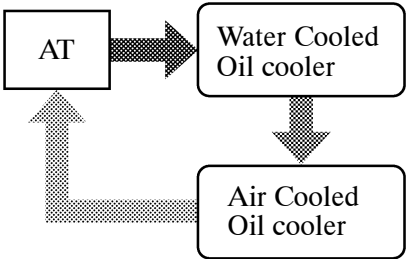
Items		Europe	China	Australia	G.C.C. Countries	General Countries
ATF Warmer		○	○	—	—	—
Oil Cooler	Water Cooled	○	○	○	○	○
	Air Cooled	—	—	○	○	○

Cooling Circuits



080CH02Y

ATF Warmer + Water Cooled Oil Cooler



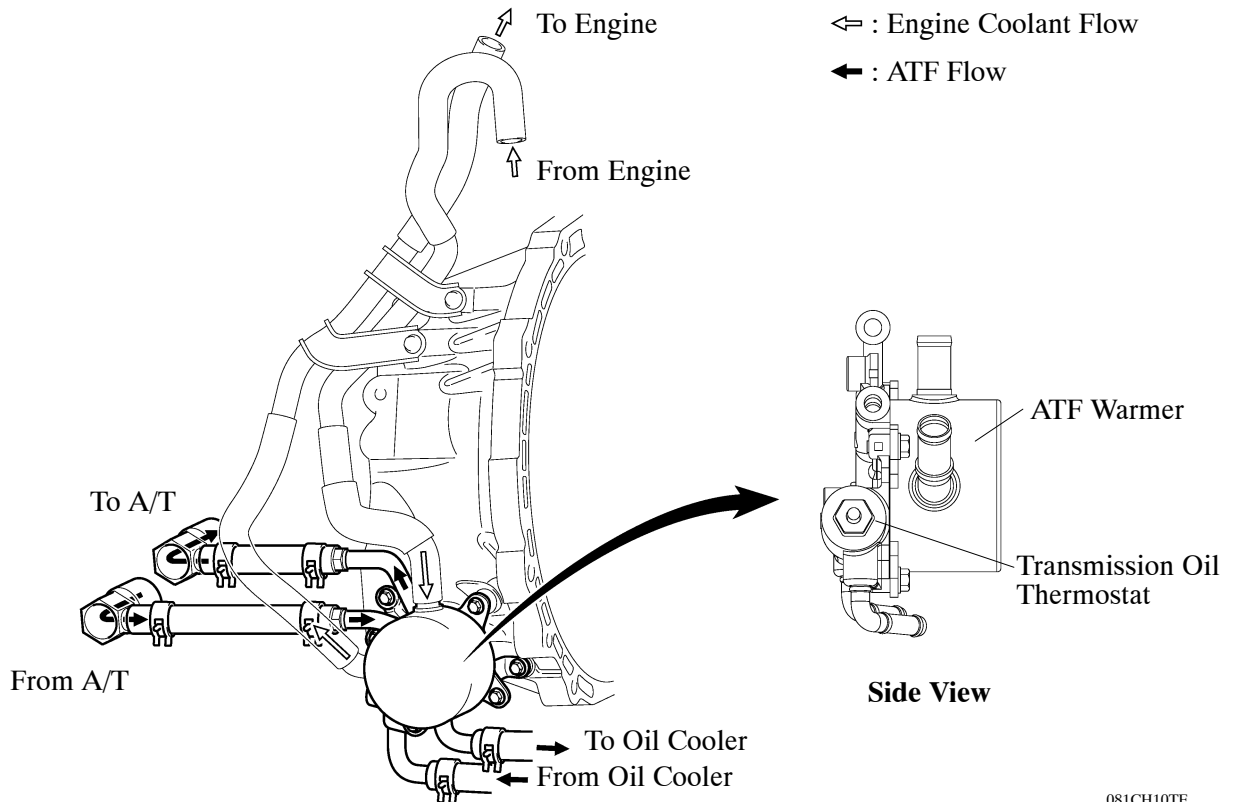
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Water Cooled Oil Cooler +Air Cooled Oil Cooler

2. ATF WARMER

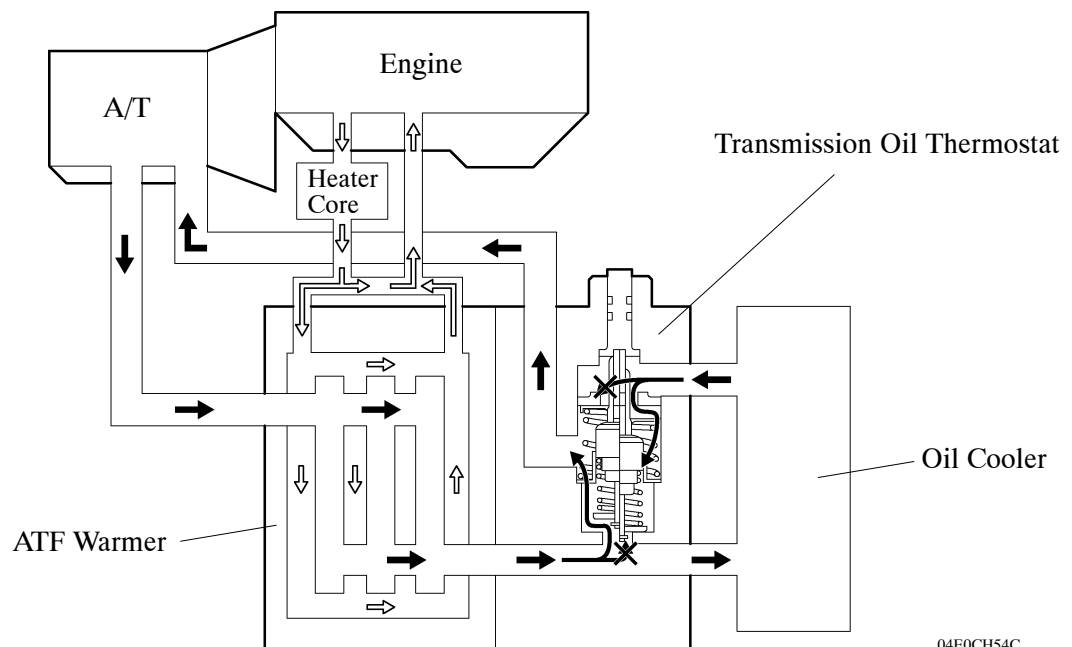
General

- The ATF warmer uses engine coolant to warm up the ATF quickly and keep the ATF temperature higher (within limits). Consequently, the friction losses of the automatic transmission are quickly reduced, thus improving fuel economy.
- The ATF warmer has a transmission oil thermostat that changes the route through which the ATF flows. When the ATF temperature is low, it is heated by the ATF warmer using the engine coolant, and when the temperature is high, it is cooled down by the ATF warmer and oil cooler.



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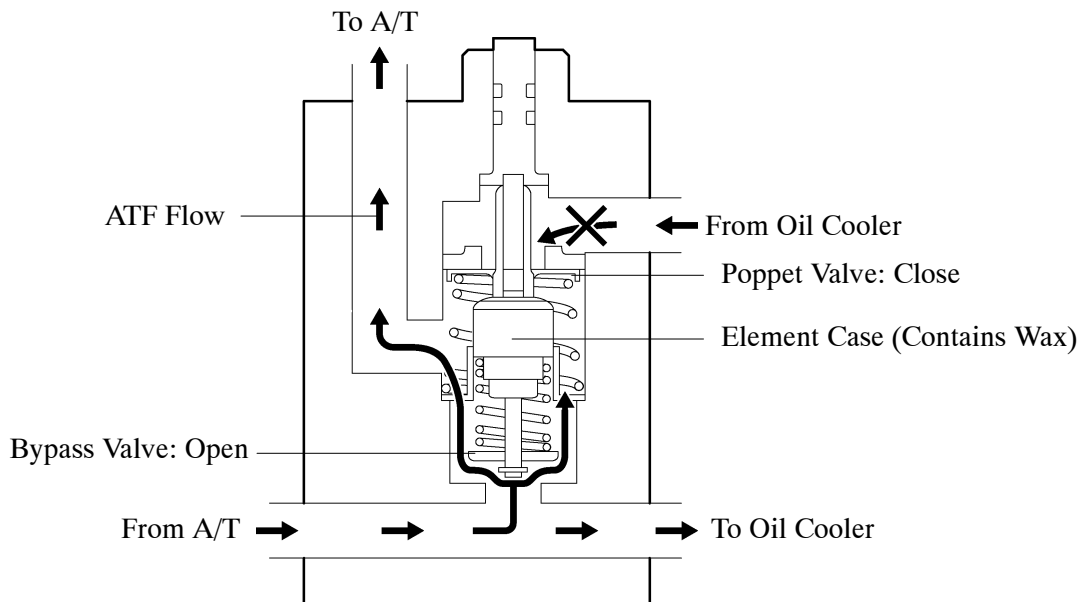
► System Diagram ◀



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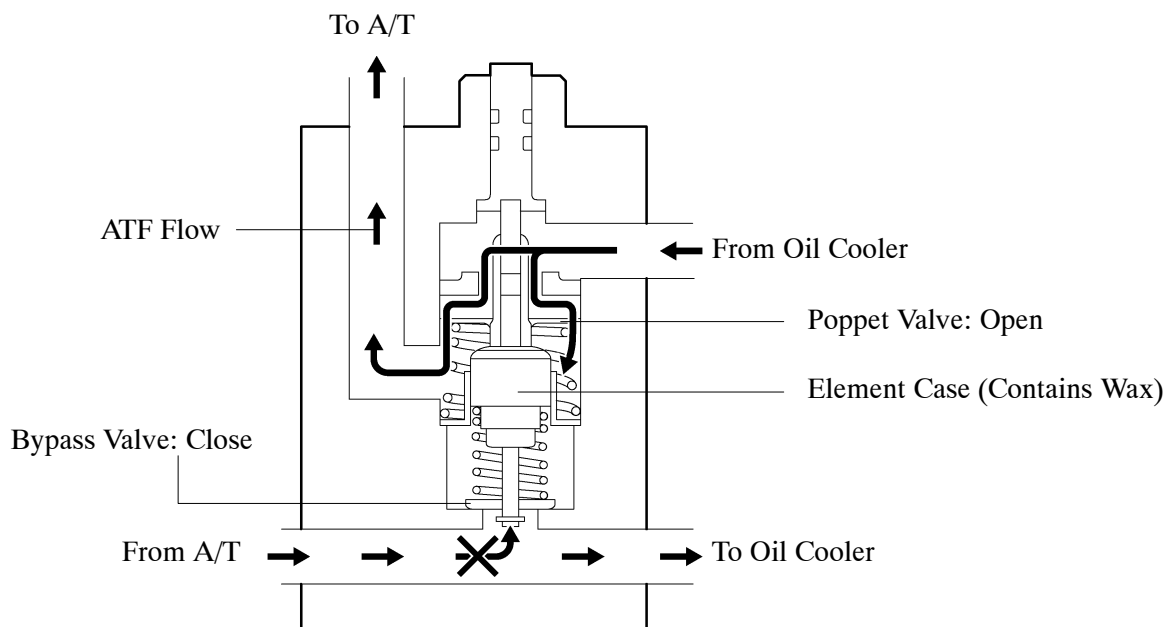
Transmission Oil Thermostat

The transmission oil thermostat consists of the poppet valve, bypass valve and element case (contains wax). When the ATF temperature changes from low to high, the wax will expand to start to open the poppet valve and close the bypass valve, thus switching the ATF passages.



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ATF Temperature: Low



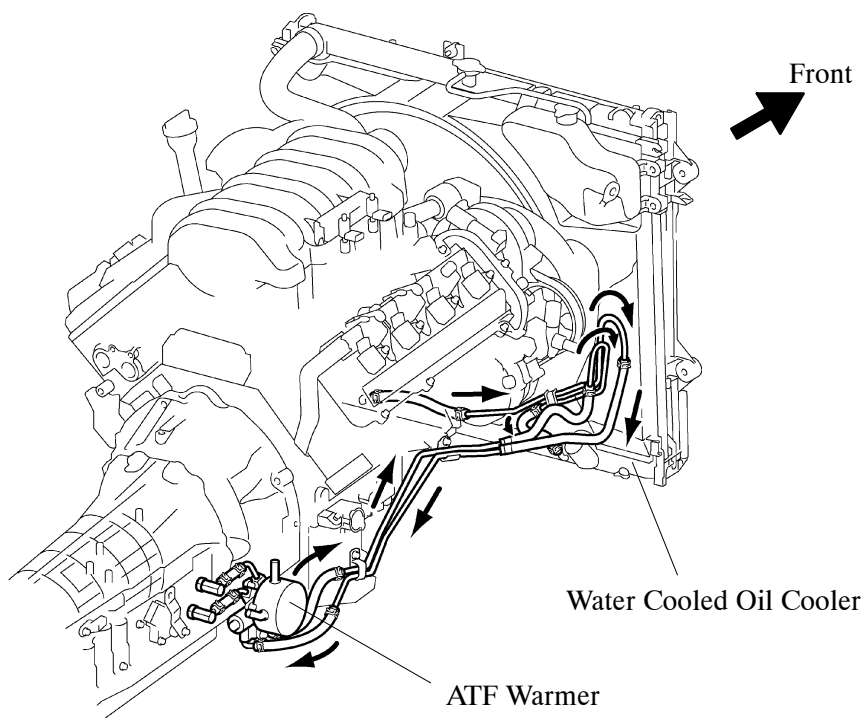
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ATF Temperature: High

3. OIL COOLER

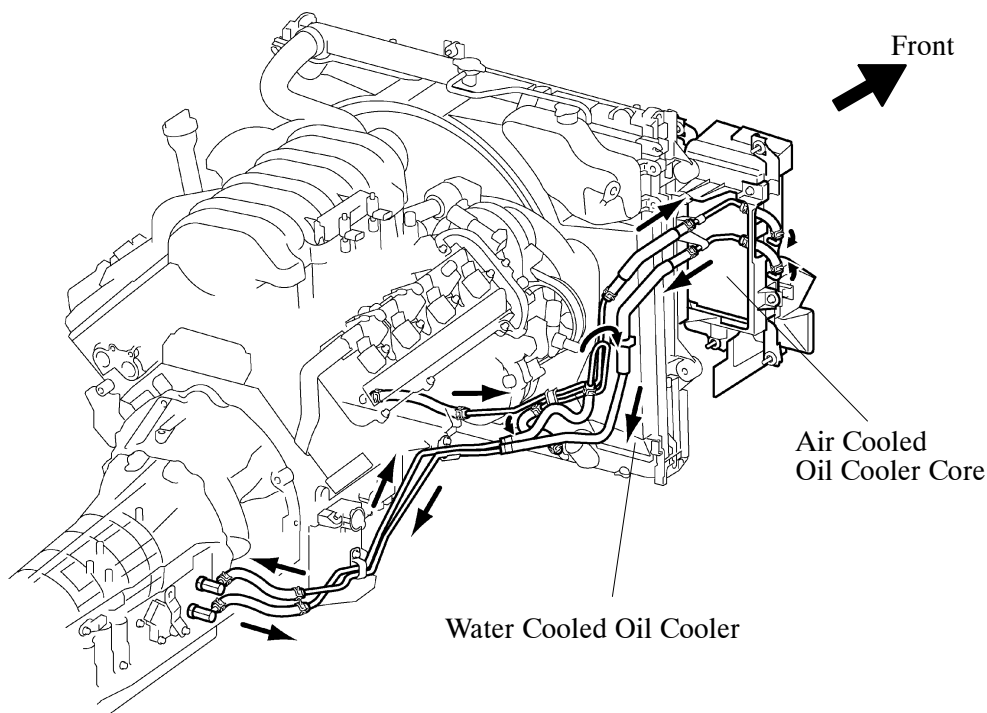
- The water cooled oil cooler cools down the ATF using the engine coolant, and is fitted on the lower of the radiator.
- The air cooled oil cooler cools down the ATF using the cooling fins which are provided on the cooler itself, and is fitted inside the radiator grille so that it can be exposed to the air directly while the vehicle is running.

► Models ATF Warmer + Water Cooled Oil Cooler ◀



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► Models Water Cooled Oil Cooler + Air Cooled Oil Cooler ◀



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